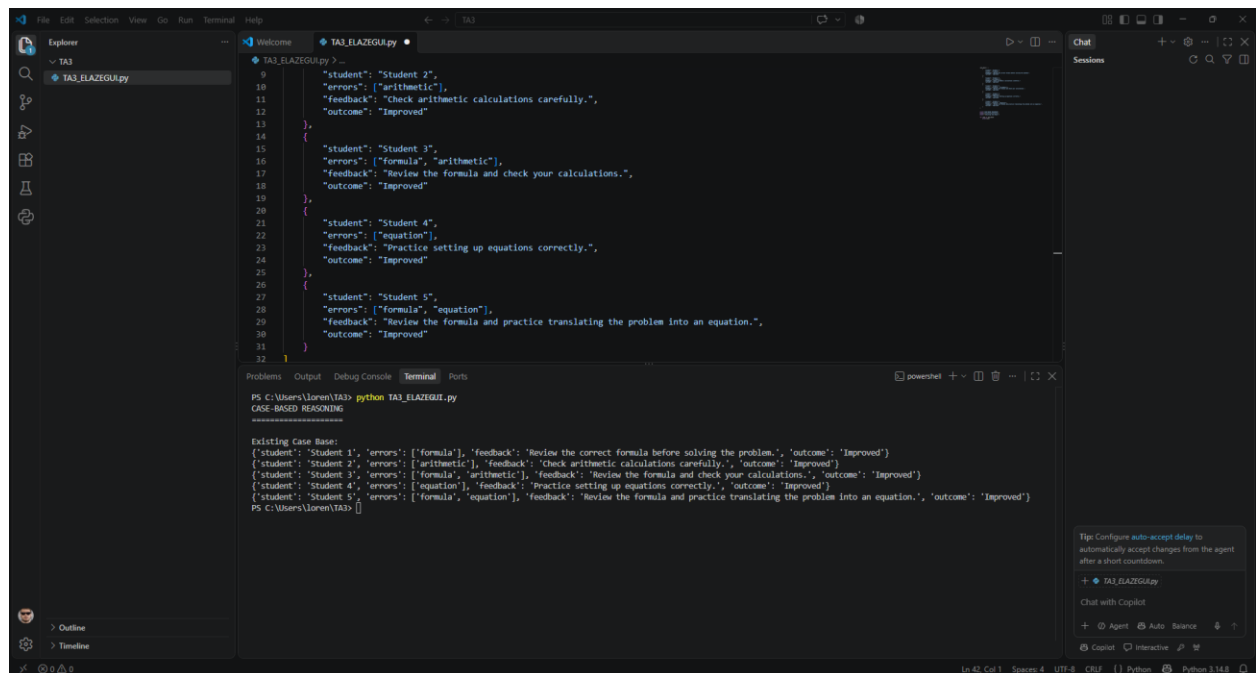


Case Study 3: Intelligent Tutoring System

A CBR system identifies the learning struggles of students. Student submits a math solution with repeated errors. System retrieves past students with similar error patterns (e.g., misusing formulas). It adapts feedback and personalized lessons based on successful interventions.

1. Case Base

The system contains five previous student cases. Each case includes the student's error patterns, feedback, and outcome. These cases serve as the knowledge base for retrieving solutions to new student problems.



```
9  "student": "Student 2",
10  "errors": ["arithmetic"],
11  "feedback": "Check arithmetic calculations carefully.",
12  "outcome": "Improved"
13  },
14  {
15  "student": "Student 3",
16  "errors": ["formula", "arithmetic"],
17  "feedback": "Review the formula and check your calculations.",
18  "outcome": "Improved"
19  },
20  {
21  "student": "Student 4",
22  "errors": ["equation"],
23  "feedback": "Practice setting up equations correctly.",
24  "outcome": "Improved"
25  },
26  {
27  "student": "Student 5",
28  "errors": ["formula", "equation"],
29  "feedback": "Review the formula and practice translating the problem into an equation.",
30  "outcome": "Improved"
31  },
32  }
```

```
PS C:\Users\loren\TAJ> python TAJ_ELAZEGUI.py
CASE-BASED REASONING
=====
Existing Case Base:
[{'student': 'Student 1', 'errors': ['formula'], 'feedback': 'Review the correct formula before solving the problem.', 'outcome': 'Improved'},
 {'student': 'Student 2', 'errors': ['arithmetic'], 'feedback': 'Check arithmetic calculations carefully.', 'outcome': 'Improved'},
 {'student': 'Student 3', 'errors': ['formula', 'arithmetic'], 'feedback': 'Review the formula and check your calculations.', 'outcome': 'Improved'},
 {'student': 'Student 4', 'errors': ['equation'], 'feedback': 'Practice setting up equations correctly.', 'outcome': 'Improved'},
 {'student': 'Student 5', 'errors': ['formula', 'equation'], 'feedback': 'Review the formula and practice translating the problem into an equation.', 'outcome': 'Improved'}]
PS C:\Users\loren\TAJ>
```

2. New Problem Description

The new student has two identified error patterns: formula and arithmetic. These errors are used by the system to compare the new problem with previous cases.

```
TAJ_ELAZIGU.py
38
39 for case in case_base:
40     print(case)
41
42
43 def calculate_similarity(new_errors, past_errors):
44     matches = 0
45
46     for error in new_errors:
47         if error in past_errors:
48             matches += 1
49
50     return matches
51
52 new_student = {
53     "Student": "New Student",
54     "errors": ["formula", "arithmetic"]
55 }
56 print("\nNew Problem:")
57 print("Student:", new_student["Student"])
58 print("Errors:", new_student["errors"])
59
```

```
Existing Case Base:
{'Student': 'Student 1', 'errors': ['formula'], 'feedback': 'Review the correct formula before solving the problem.', 'outcome': 'Improved'}
{'Student': 'Student 2', 'errors': ['arithmetic'], 'feedback': 'Check arithmetic calculations carefully.', 'outcome': 'Improved'}
{'Student': 'Student 3', 'errors': ['formula', 'arithmetic'], 'feedback': 'Review the formula and check your calculations.', 'outcome': 'Improved'}
{'Student': 'Student 4', 'errors': ['equation'], 'feedback': 'Practice setting up equations correctly.', 'outcome': 'Improved'}
{'Student': 'Student 5', 'errors': ['formula', 'equation'], 'feedback': 'Review the formula and practice translating the problem into an equation.', 'outcome': 'Improved'}
PS C:\Users\loren\TAJ> python TAJ_ELAZIGU.py
CASE-BASED REASONING
Existing Case Base:
{'Student': 'Student 1', 'errors': ['formula'], 'feedback': 'Review the correct formula before solving the problem.', 'outcome': 'Improved'}
{'Student': 'Student 2', 'errors': ['arithmetic'], 'feedback': 'Check arithmetic calculations carefully.', 'outcome': 'Improved'}
{'Student': 'Student 3', 'errors': ['formula', 'arithmetic'], 'feedback': 'Review the formula and check your calculations.', 'outcome': 'Improved'}
{'Student': 'Student 4', 'errors': ['equation'], 'feedback': 'Practice setting up equations correctly.', 'outcome': 'Improved'}
{'Student': 'Student 5', 'errors': ['formula', 'equation'], 'feedback': 'Review the formula and practice translating the problem into an equation.', 'outcome': 'Improved'}
New Problem:
Student: New Student
Errors: ['formula', 'arithmetic']
PS C:\Users\loren\TAJ>
```

3. Similarity Assessment and Retrieved Case

The system compares the new student's errors with the errors from each previous case. A similarity score is calculated based on the number of matching errors. The case with the highest score is retrieved as the most similar previous case.

```
TAJ_ELAZIGU.py
65 score = calculate_similarity(
66     new_student["errors"],
67     case["errors"]
68 )
69
70 print(
71     case["Student"],
72     "Similarity Score:",
73     score
74 )
75
76 if score > best_score:
77     best_score = score
78     best_case = case
79     print("\nRetrieved Similar Case:")
80     print("Student:", best_case["Student"])
81     print("Similarity Score:", best_score)
82     print("Previous Feedback:", best_case["feedback"])
83
84
85
86
```

```
{'Student': 'Student 2', 'errors': ['arithmetic'], 'feedback': 'Check arithmetic calculations carefully.', 'outcome': 'Improved'}
{'Student': 'Student 3', 'errors': ['formula', 'arithmetic'], 'feedback': 'Review the formula and check your calculations.', 'outcome': 'Improved'}
{'Student': 'Student 4', 'errors': ['equation'], 'feedback': 'Practice setting up equations correctly.', 'outcome': 'Improved'}
{'Student': 'Student 5', 'errors': ['formula', 'equation'], 'feedback': 'Review the formula and practice translating the problem into an equation.', 'outcome': 'Improved'}
New Problem:
Student: New Student
Errors: ['formula', 'arithmetic']
Similarity Assessment:
Student 1 Similarity Score: 1
Retrieved Similar Case:
Student 2 Similarity Score: 1
Student 3 Similarity Score: 2
Retrieved Similar Case:
Student 4 Similarity Score: 0
Student 5 Similarity Score: 1
Student: Student 3
Similarity Score: 2
Previous Feedback: Review the formula and check your calculations.
PS C:\Users\loren\TAJ>
```

4. Adapted Final Solution

```
def adapt_solution(errors):
    if "arithmetic" in errors:
        feedback.append("Check your arithmetic calculations carefully.")
    if "equation" in errors:
        feedback.append("Practice setting up equations correctly.")
    return ".join(feedback)
final_feedback = adapt_solution(
    new_student["errors"]
)
print("\nAdapted Final Solution:")
print(final_feedback)
```

```
{'student': 'Student 5', 'errors': ['formula', 'equation'], 'feedback': 'Review the formula and practice translating the problem into an equation.', 'outcome': 'Improved'}

New Problem:
Student: New Student
Errors: ['formula', 'arithmetic']

Similarity Assessment:
Student 1 Similarity Score: 1

Retrieved Similar Case:
Student 2 Similarity Score: 1
Student 3 Similarity Score: 2

Retrieved Similar Case:
Student 4 Similarity Score: 0
Student 5 Similarity Score: 1
Student: Student 3
Similarity Score: 2
Previous Feedback: Review the formula and check your calculations.

Adapted Final Solution:
Review the correct formula before solving. Check your arithmetic calculations carefully.
PS C:\Users\loren\IA3>
```

5. Retention / Updated Case Base

After generating the final feedback, the new student case is added to the case base.

This demonstrates the retention stage of Case-Based Reasoning because the system stores the newly solved case for future use.

```
Review the correct formula before solving. Check your arithmetic calculations carefully.
PS C:\Users\loren\IA3> python TAJ_ELAZGUI.py

CASE-BASED REASONING

Existing Case Base:
{'student': 'Student 1', 'errors': ['formula'], 'feedback': 'Review the correct formula before solving the problem.', 'outcome': 'Improved'}
{'student': 'Student 2', 'errors': ['arithmetic'], 'feedback': 'Check arithmetic calculations carefully.', 'outcome': 'Improved'}
{'student': 'Student 3', 'errors': ['formula', 'arithmetic'], 'feedback': 'Review the formula and check your calculations.', 'outcome': 'Improved'}
{'student': 'Student 4', 'errors': ['equation'], 'feedback': 'Practice setting up equations correctly.', 'outcome': 'Improved'}
{'student': 'Student 5', 'errors': ['formula', 'equation'], 'feedback': 'Review the formula and practice translating the problem into an equation.', 'outcome': 'Improved'}

New Problem:
Student: New Student
Errors: ['formula', 'arithmetic']

Similarity Assessment:
Student 1 Similarity Score: 1

Retrieved Similar Case:
Student 2 Similarity Score: 1
Student 3 Similarity Score: 2

Retrieved Similar Case:
Student 4 Similarity Score: 0
Student 5 Similarity Score: 1
Student: Student 3
Similarity Score: 2
Previous Feedback: Review the formula and check your calculations.

Adapted Final Solution:
Review the correct formula before solving. Check your arithmetic calculations carefully.

New case saved successfully.

Updated Case Base:
{'student': 'Student 1', 'errors': ['formula'], 'feedback': 'Review the correct formula before solving the problem.', 'outcome': 'Improved'}
{'student': 'Student 2', 'errors': ['arithmetic'], 'feedback': 'Check arithmetic calculations carefully.', 'outcome': 'Improved'}
{'student': 'Student 3', 'errors': ['formula', 'arithmetic'], 'feedback': 'Review the formula and check your calculations.', 'outcome': 'Improved'}
{'student': 'Student 4', 'errors': ['equation'], 'feedback': 'Practice setting up equations correctly.', 'outcome': 'Improved'}
{'student': 'Student 5', 'errors': ['formula', 'equation'], 'feedback': 'Review the formula and practice translating the problem into an equation.', 'outcome': 'Improved'}
{'student': 'New Student', 'errors': ['formula', 'arithmetic'], 'feedback': 'Review the correct formula before solving. Check your arithmetic calculations carefully.', 'outcome': 'Pending'}
PS C:\Users\loren\IA3>
```